

LLM Context Document

AI-Powered Sales Forecasting Dashboard

Project Goal

Build an end-to-end BI portfolio project using SQL, Power BI, Python forecasting and Streamlit. Deliverables include star schema, SQL analytics, 5-page Power BI dashboard, forecasting, documentation and GitHub-ready assets.

Dataset

Global Superstore dataset (~51K records). Imported: sales_clean.csv (validation), fact_sales.csv, dim_customer.csv, dim_product.csv, dim_region.csv, dim_date.csv.

Data Model

Star Schema. Relationships: dim_customer→fact_sales(customer_key), dim_product→fact_sales(product_key), dim_region→fact_sales(region_key), dim_date→fact_sales(date_key). Date table marked using dim_date[full_date]. Created disconnected _KeyMeasures table for measures.

Power BI Theme

Corporate light theme. Primary #003B73, Secondary #0074B7, Success #27AE60, Alert #BF212F, Background #F8F9FA, Segoe UI font, rounded cards, subtle shadows.

Core DAX

Revenue, Profit, Orders, Quantity Sold, Customer Count, Profit Margin %, Avg Order Value, Avg Discount, Revenue per Customer, PY Revenue, YoY Growth %, YTD/MTD/QTD Revenue, Customer/Regional/Product measures.

Dashboard Pages

1. Executive Summary: KPIs, Revenue-Profit trend, Market revenue, Category revenue. 2. Product & Sales: Product ranking, category matrix, profitability, discounts. 3. Customer Insights: Segments, top customers, repeat buyers, churn indicators. 4. Regional Performance: Markets, countries, cities, regional matrix. 5. Forecasting & Advanced Analytics: Forecast, anomalies, key influencers, decomposition tree.

Design Decisions

Used navigation buttons, synchronized slicers (Year, Quarter, Market), custom subtitles, info icon, KPI cards, insight cards, conditional formatting, professional spacing. Recommended Treemap instead of repeated maps where appropriate.

Known Issues & Resolutions

- Latest Year Revenue blank due to dim_date extending to 2015; use latest sales year from fact table.
- Card visuals don't support UNICHAR line breaks reliably.
- Gradient fill inside bar charts not supported; use field-value colors or corporate palette.
- Forecast option depends on line chart with continuous date axis and Power BI version.

Folder Structure

01_Business_Understanding, 02_Dataset, 03_SQL, 04_PowerBI, 05_Python_Forecasting, 06_Streamlit, 07_Documentation, 08_GitHub.

Deliverables

PBIX dashboard, SQL scripts, DAX documentation, dashboard report, GitHub README, forecasting notebook, final portfolio documentation.

Purpose of this Document

This condensed context captures the key architectural decisions, datasets, measures, dashboard design, and implementation choices made during development. It is intended to be supplied to another LLM so it can continue the project without requiring the entire chat history, minimizing token usage while preserving essential context.